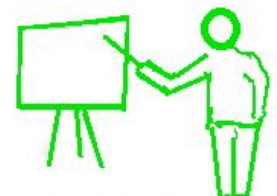




Agile Development Awareness session



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Section

1

Agile Vs Traditional

Agile Manifesto

profile of authors at <http://www.agilemanifesto.org/authors.html>

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Manifesto for Agile development

We are covering better ways of delivering software by doing it and helping others do it.

Through this work we have come to value:

Individuals & Interactions over processes and tools

Working software over comprehensive Documentation

Customer Collaboration over contract negotiation

Responding to Change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

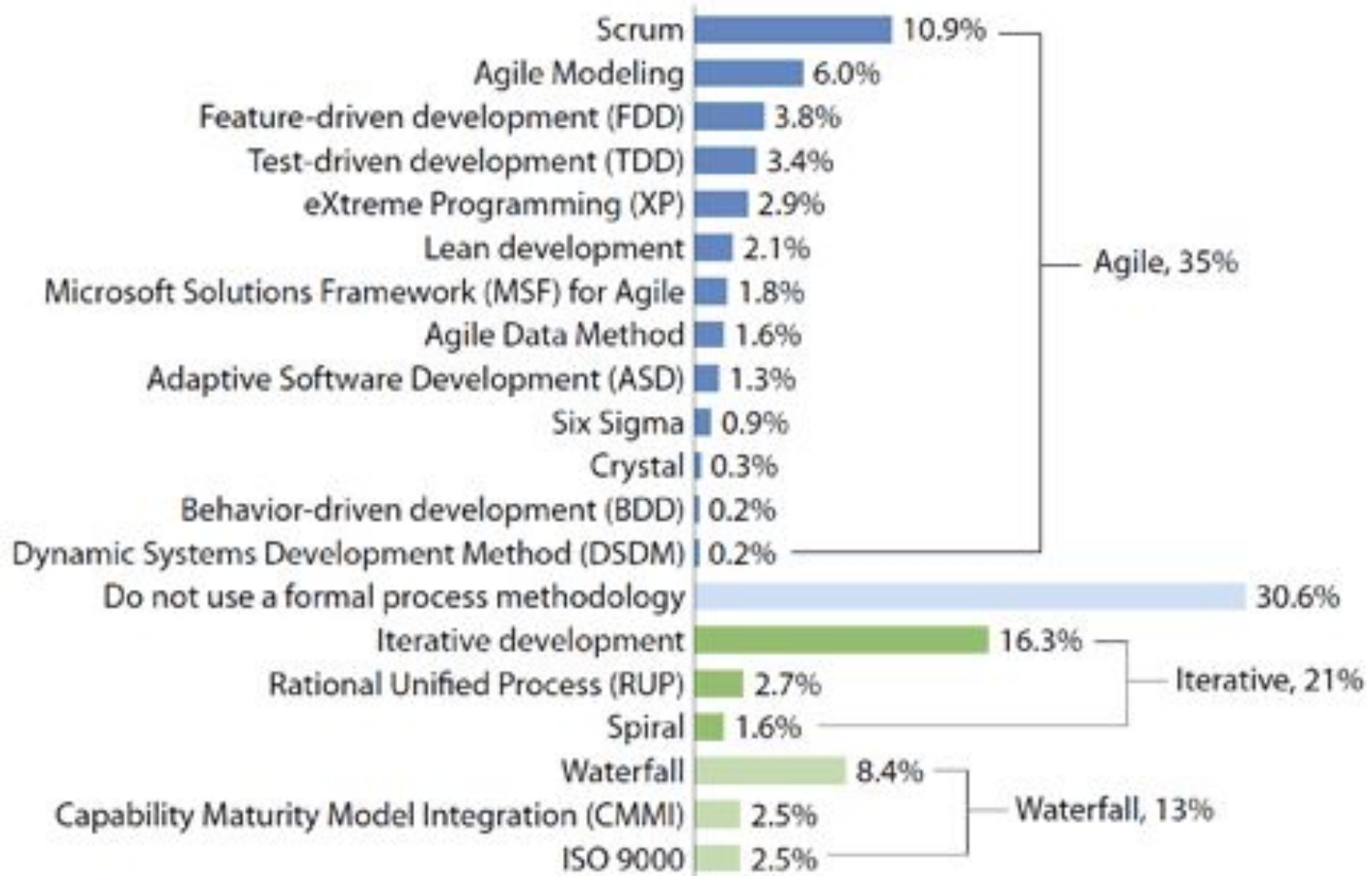
Kent Beck
Mike Beedle
Arie Van Bennekum
Alistair Cockburn
Warid Cuningham
Martin Fowler

James Grenning
Jim Highsmith
Andrew Hunt
Ron Jeffries
Jon Kern
Brian Marick

Robert C. Martin
Steve Mellor
Ken Schwaber
Jeff Sutherland
Dave Thomas

Technology Share

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Source: <http://www.spkaa.com/engineering-technology-services/software-engineering/workflow-process-definition>

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What will be covered– 180⁰ Shifts

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1. **Team**
2. **Requirements**
3. **Development iteration**
4. **Acceptance Tests**
5. **Programming Style**
6. **Testing**
7. **Ownership**
8. **Integration**
9. **Development Pace**
10. **Workspace**
11. **Design & architecture**
12. **Refactoring**
13. **Documentation**
14. **Task allocation**
15. **Progress meeting**
16. **Status meeting**
17. **Project status**

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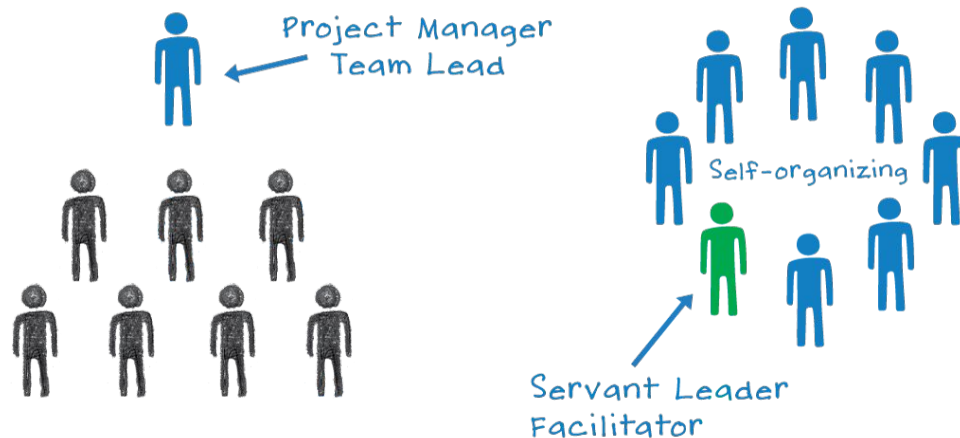
Team

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Team means, analyst, designer, coder, tester, architect, team lead, QA manager, and project manager , in hieratical order

**In agile development we believe in
Whole Team**

**in an agile project, the customer, however defined, is a member
of, and available to, the team.**



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Requirements

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Requirements are captured and written by analyst , and project manager and then verified from customer



In agile development we believe in

User stories by customers

the customer writes a few words on an index card .

Development iteration

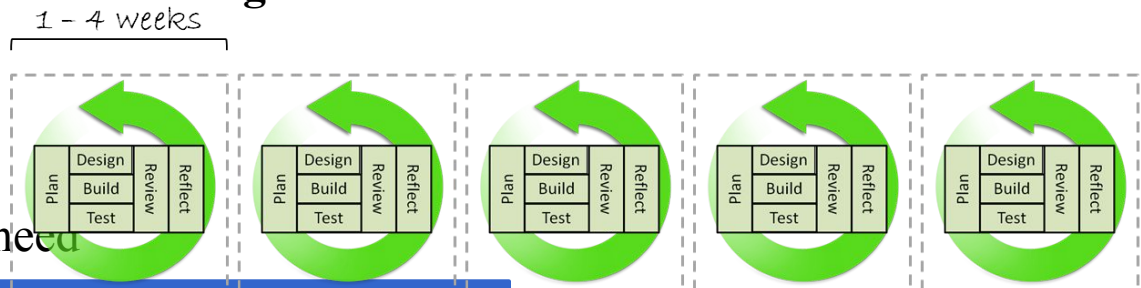
8

Projects are developed in iterations of 2-3 months, and first iteration act as detailed design and plan of the whole project, and remaining are development iterations

**In agile development we believe in
Short cycle**

An agile project delivers working software every two to four weeks. Each of these iterations produces working software that addresses some of the needs of the stakeholders.

At the end of each iteration, the system is demonstrated to the stakeholders in order to get their feedback.



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Acceptance Tests

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Requirements that are captured at initial level and contracted are used as an acceptance test at the end

In agile development we believe in

Acceptance tests are written by business analysts, quality assurance specialists, and testers during the iteration.

The language they are written in is easy for programmers, customers, and businesspeople to read and understand.

Once an acceptance test passes, it is added to the body of passing acceptance tests and is never allowed to fail again



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Programming Style

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Each coder is assigned task by the team lead or Project manager, then QA team check for the errors

In agile development we believe in

Pair Programming

In agile Code is written by pairs of programmers working together at the same workstation. One member of each pair drives the keyboard and types the code. The other member of the pair watches the code being typed, finding errors and improvements.^[2] The two interact intensely. Both are completely engaged in the act of writing software.

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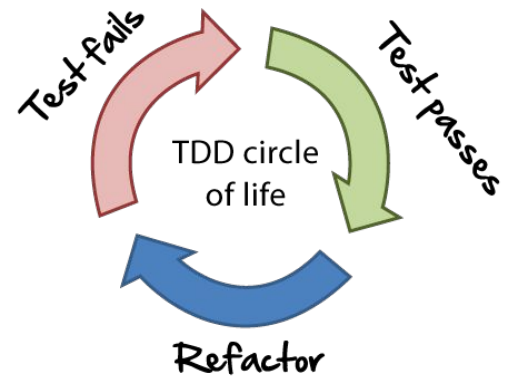
Testing

11

Usually test cases are developed at the time of prototype approval, then code is written by team and QA team tests the code for errors

In agile we use Test Driven Development (TDD)

in which ,all production code is written in order to make a failing unit test pass. **First, we write a unit test that fails** because the functionality it is testing for doesn't exist. **Then we write the code that makes that test pass.**



Ownership

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manager is whole sole responsible for the whole project

**In agile development we believe in
Collective ownership**

No team member is individually responsible for any one particular module or technology. Everybody works on the graphical user interface (GUI). Everybody works on the middleware. Everybody works on the database. Nobody has more authority than anybody else over a module or a



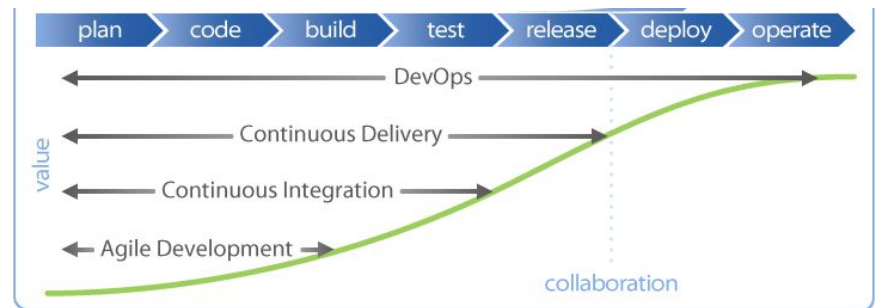
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Integration

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Software integration is done at the end of module and/or end of project , and is done by the QA team

**I agile w believe in
Continuous integration**



The programmers check their code in and integrate several times per day. The rule is simple. The first one to check in wins; everybody else merges.

So Agile teams will build the system many times each day. They build the whole system from end to end.

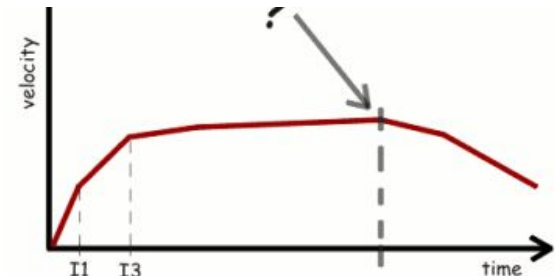
Development Pace

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In software engineering, we use 80/20 principle,
in initial 20%, we develop in high speed and recover 50% of amount,
and remaining 60% in moderate speed,
and in last 20% we speed up one again

**I agile we use
Sustainable Pace**

**A software project is not a sprint; it is a marathon.
The rule is that a team is not allowed to work overtime.**

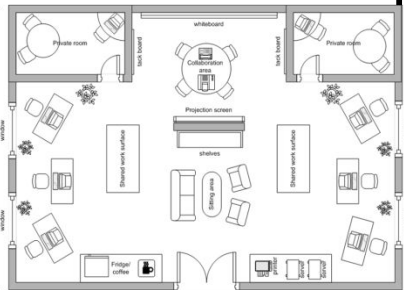


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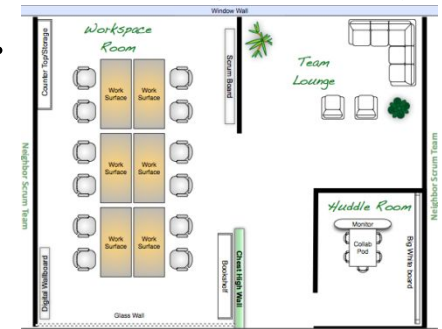
Workspace

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Most of software houses have separate room/cubical of each team member, and usually tasks are assigned by the PM or via software.



Agile believes in open workspace



The team works together in an open room. Tables are set up with workstations on them. Each table has two or three such workstations. Two chairs are in front of each workstation. The walls are covered with status charts, task breakdowns, Unified Modeling Language (UML) diagrams, and so on.

Design and Architecture

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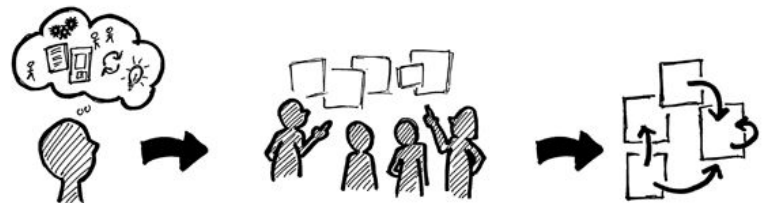
We design and architect the Whole project at the beginning .

In agile we believe in Simple Design

An agile team makes its designs as simple and expressive as they can be.

Furthermore, the team narrows its focus to consider only the stories that are planned for the **current iteration**, not worrying about stories to come.

Rather, the team migrates the design of the system from iteration to iteration to be the best design for the stories that the system currently implements.



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· ENVISION ·

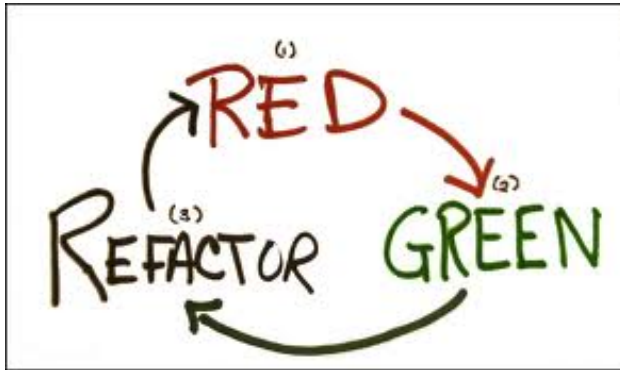
· FACILITATE ·

· PROTOTYPE ·

Refactoring

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Refactoring is the practice of making a series of tiny transformations that improve the structure of the system without affecting its behavior. **And** usually done at the end of project and initiated by QA team



**In agile we believe in
Continuous Refactoring**

Agile teams reverse this degradation through frequent refactoring. Each transformation is trivial, hardly worth doing. But together, they combine into significant transformations of the design and architecture of the system.

Refactoring is something we do every hour or every half hour. Through refactoring, we continuously keep the code as clean, simple, and expressive as it can be.

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- [illegible]

Agile Documentation

Vs.

Waterfall Documentation

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Task assignment

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Project manager/team lead assigns task to whole team

In agile we believe in Task selection

*the team decides/selects how much work they will commit to complete, rather than having it assigned to them by the Product Owner.
This makes for a much more reliable*

Backlog Item	Task	Owner
Upgrade transaction processing module	Configure database and space IDs for Trac	Sanjay
	Use test data to tune the learning and action model	Jing
	Setup a cart server code to run as apache server	Philip
	Implement pre-Login Handler	Tracy
Investigate solutions for speeding up credit card validation	Merge DCP code and complete layer-level tests	Jing
	Complete machine order for pRank	Jing
	Change DCP and reader to use pRank http API	Tracy

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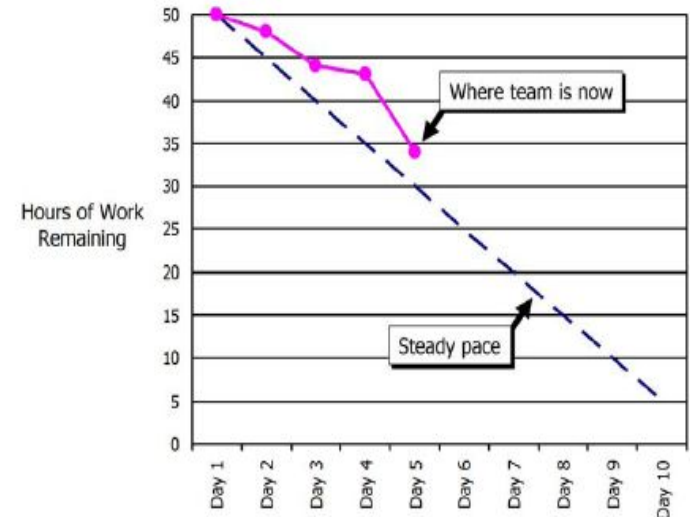
Progress monitoring

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Progress of project is monitored by defining how many and how much tasks are developed

In agile we believe in Burndown Chart

- It shows, each day, how much work (measured in hours or days) **remains** until the team's commitment is completed.



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Status meeting



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Progress of project is monitored/demanded PM in weekly meetings. And every one reports to PM

In agile we believe in Daily stand up meeting

One by one, each member of the team reports just three things to the other members of the team:

1. What they were able to get done since the last meeting.
2. what they're aiming to get done by the next meeting.
3. any blocks or obstacles that are in their way.

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Status of project

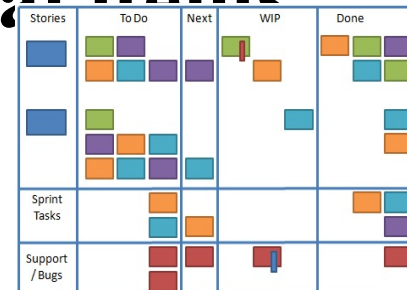
22

Status of project is shown by the no of tasks that are completed

In agile we believe in Status board

Status is shown on two **sheets of poster-sized paper** labeled “What’s Working Well” and “What’s Not Working, or Could Work Better” – and then have **each person add several items to either.**

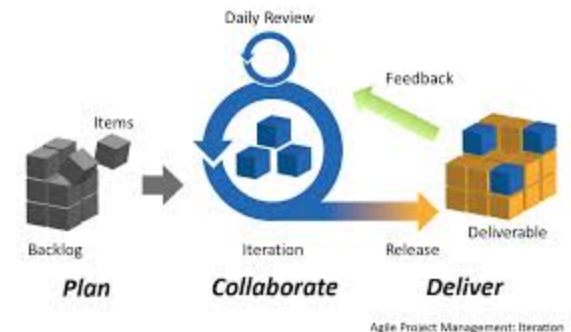
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Agile Development

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- Agile emphasizes building working software that people can get hands on with quickly, versus spending a lot of time writing specifications up front.
- Agile focuses on small, cross-functional teams empowered to make decisions, versus big hierarchies and compartmentalization by function,.
- Agile focuses on rapid iteration, with as much customer input along the way as possible.

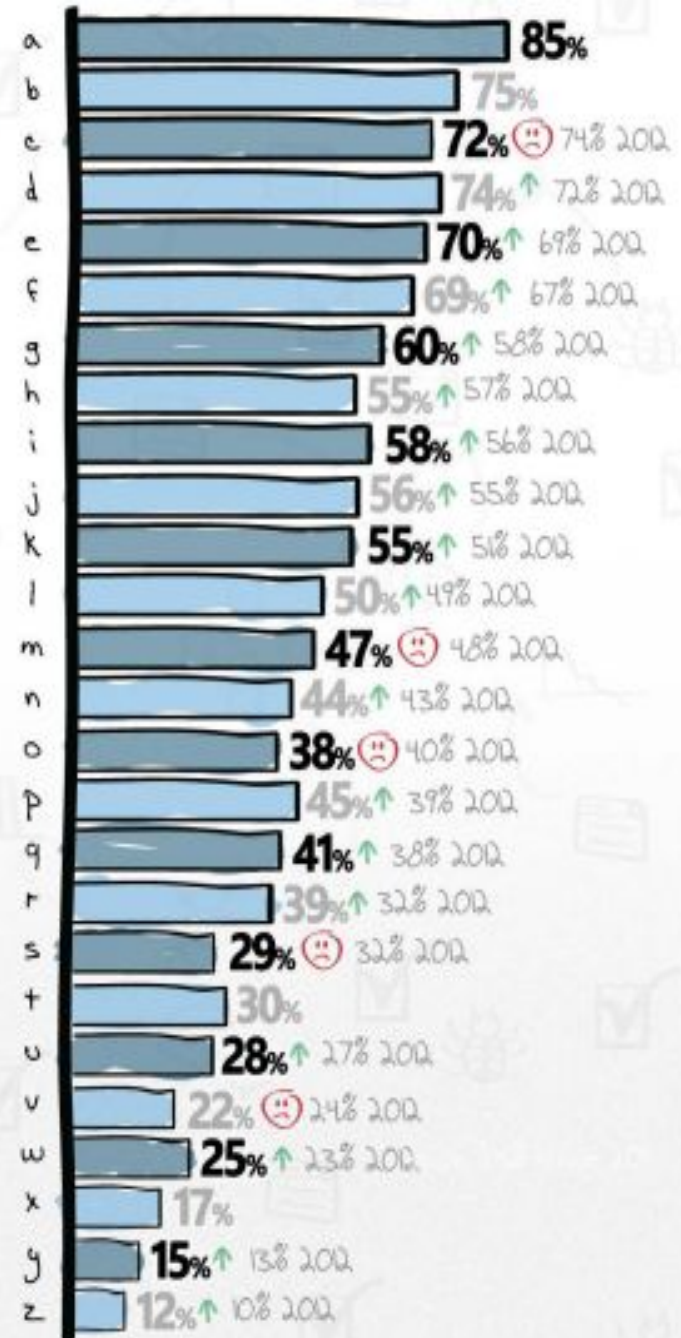


AGILE TECHNIQUES EMPLOYED

Respondents are making use of a wide variety of different agile management techniques. More than 85% practice Daily Standups, ¾ are using Iteration Planning and Retrospectives, and nearly the same proportion said they maintain Burndown charts. Over the last 2 years we've seen a 10% increase in the use of Retrospectives (from 64% in 2011 to 74% in 2013).

*Respondents were able to select multiple options.

- | | |
|-----------------------------------|--------------------------------|
| a Daily Standup | n Open Workarea |
| b Iteration Planning | o TDD |
| c Unit Testing | p Digital Taskboard |
| d Retrospectives | q Story Mapping |
| e Release Planning | r Kanban |
| f Burndown/ Team-Based Estimation | s Collective Code Ownership |
| g Velocity | t Pair Programming |
| h Coding Standards | u Automated Acceptance Testing |
| i Continuous Integration | v Analog Taskboard |
| j Automated Builds | w Continuous Deployment |
| k Dedicated Product Owner | x Agile Games |
| l Integrated Dev/QA | y Cycle Time |
| m Refactoring | z BDD |



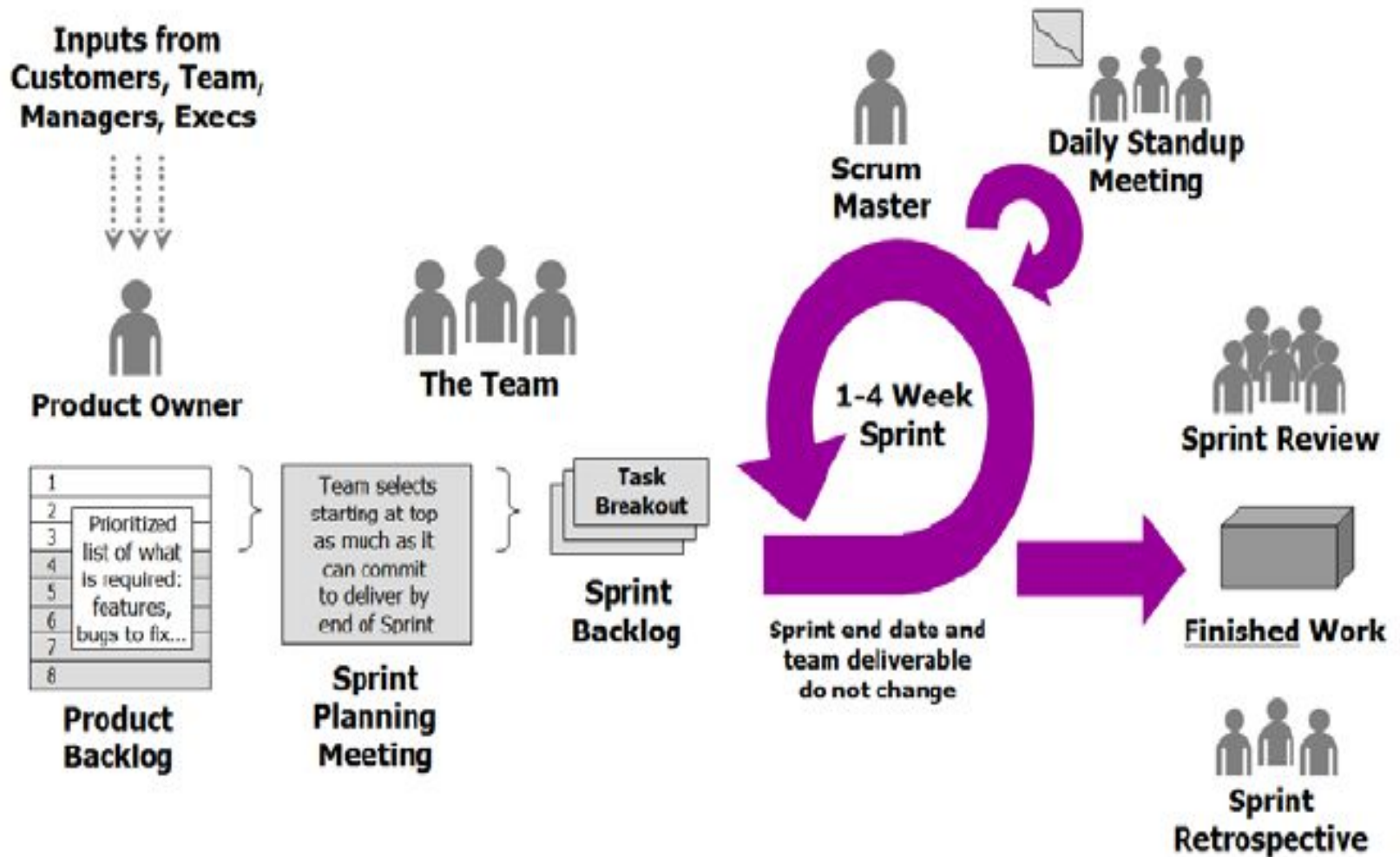


Figure 1. Scrum

Prepared and presented by Usman

Waheed

Thanks

Q/A